

A Subfield Subcode Construction of a Linear Exact Repair Scheme

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Abstract

In a *linear exact repair scheme (LERS)*, a single codeword symbol can be efficiently recovered from subfield symbols by using the remaining symbols. The existence of a LERS follows from the existence of a particular set of dual codewords. Exploiting this implication, we present a randomized construction of a LERS that achieves a repair bandwidth that does not exceed $n - 1$, where n denotes the length of the code. Our scheme applies to general codes of length n and dimension k over a finite field of size q^t satisfying $k \leq (n - 1)/t - 1$, thereby covering a new range of parameters.

Key Words : Subfield subcode, erasure recovery, linear repair scheme, distributed storage.

1 Introduction

1.1 Motivation

As advances in technology enable ever more powerful mechanisms for data acquisition and curation, distributed storage systems have become increasingly important. In large-scale infrastructures, the failure of a single code symbol is a common event in the use of distributed storage systems [PWB07, DGW⁺10]. Moreover, since such systems are constrained by storage and communication costs, the efficient repair of erased code symbols has become a central challenge. The repair problem—namely, recovering a failed code symbol using the remaining symbols—was first formalized in [DGW⁺10], where the authors also established a lower bound on the exact repair bandwidth for general MDS codes. Here, the repair bandwidth b denotes the number of base-field elements that must be downloaded from d other nodes in order to recover a single failed code symbol. We state a simplified form of this lower bound below.

Theorem 1 (Cut-set bound [DGW⁺10]). *Let C be an MDS code over $\mathbb{F}_{q^t}$ of length n and dimension k . Then the repair bandwidth b for a single erasure satisfies*

$$b \geq \frac{dt}{d - k + 1},$$

where d , with $k < d < n$, denotes the number of contacted code symbols.

Note that for $d = n - 1$, this reduces to the cut-set bound $b \geq t \cdot (n - 1)/(n - k)$. This cut-set bound is only achievable for the parameter regime where $t \gg n - k$; specifically, it has been shown that $t \geq k^2$ is required. In contrast, in the regime where $t \ll n - k$, the trivial lower bound $b \geq k + t - 1$ holds¹. Note that in this setting, the trivial lower bound is indeed larger than the cut-set bound. The scheme proposed in [GW16] is optimal among linear repair schemes for full-length Reed–Solomon codes with $n = q^t$. Here, $t = \log_q(n)$ is small compared to $n - k$ and therefore the trivial lower bound applies. In particular, their LERS based on trace polynomials reconstructs an erasure by downloading only partial information from each of the $n - 1$ surviving symbols. Compared with Lagrange interpolation, this yields a smaller repair bandwidth in certain parameter regimes, namely when $n = q^t$ and $k \leq n - q^{t-1}$. This phenomenon is also striking from a secret sharing perspective. Due to the close relationship between Reed–Solomon codes and Shamir secret sharing, these repair (or reconstruction) procedures may be viewed as communication-efficient reconstruction schemes that still guarantee perfect secrecy against any adversary observing fewer than k shares. Moreover, Guruswami and Wootters [GW16], together with Dau and Milenkovic [DM17], generalized the repair bandwidth for Reed–Solomon codes to at most $(n - 1)(t - s)$ when the smaller field is chosen to be $\mathbb{F}_{q^{t-s}}$, under the condition $n - k \geq q^s$, for an integer $0 < s < t$. Building on this approach, a repair scheme for Reed–Solomon codes based on subspace polynomials was constructed in [DXK⁺21], thereby broadening the range of parameters for which optimal repair bandwidth can be achieved. Furthermore, it is shown in [YB17] that the framework of [GW16] can be used to meet the cut-set bound for Reed–Solomon codes with parameters $n \leq q^t$ and $t = (n - k)^n$, achieving a repair bandwidth of $t \cdot (n - 1)/(n - k)$. Multiple-erasure scenarios were studied in [DXK⁺21, DDKM17, MBW17]. Finally, [CT22] presents a nonlinear repair scheme for Reed–Solomon codes over prime fields. These results, together with the wide range of real-world applications of Reed–Solomon codes (e.g., Google File System [Fik] and HDFS-RAID [Bor10]), underline the importance of further investigating the reconstruction of erased code symbols.

¹In an MDS code, any $k - 1$ symbols do not determine the remaining symbol; thus, even if one sub-symbol from $k - 1$ different symbols is given, at least the information of one additional full symbol, i.e., t sub-symbols, is necessary. Hence, the total download is at least $k + t - 1$.

1.2 Contributions

In this work, we study the repair problem for non-degenerate general codes (see Definition 7). More precisely, our main contribution is a novel construction of a linear exact repair scheme via subfield subcodes. We develop a LERS for any non-degenerate code of length n and dimension k over an extension field of degree t of $\mathbb{F}_q$, provided that $k \leq (n-1)/t - 1$. This construction yields a LERS for parameter ranges not covered by previously known schemes, in particular for large values of q and t .

1.3 Organization

Section 2 introduces the notation, definitions, and basic results used throughout the paper. In Section 3, we present the construction of a novel LERS via subfield subcodes. Section 4 concludes the paper.

2 Preliminaries

2.1 Shortening and puncturing

We recall standard properties of Reed–Solomon codes and their generalizations. Most of this material can be found in [MS77, Cou20]. Let $\mathbb{B} = \mathbb{F}_q$ be the finite field of size q , and let $\mathbb{F}$ be an extension field of $\mathbb{B}$ of degree t . We consider codes of length n over the alphabet $\mathbb{F}$, and use the notation $[n, k]$ to denote a code of length n and dimension k ². We denote the dual of a code C by $C^\perp$. When the alphabet needs to be specified, we write $[n, k]_{\mathbb{F}}$ or $[n, k]_{\mathbb{B}}$. Let C be an $[n, k]$ linear code over $\mathbb{F}$, and let $I \subseteq \{0, \dots, n-1\}$ be an index set. The *puncturing* of C at I , denoted by $\underline{C}_I$, is the code of length $n - |I|$ obtained by deleting the coordinates indexed by I from every codeword of C . Formally, $\underline{C}_I := \{(c_j)_{j \in \{0, \dots, n-1\} \setminus I} \mid (c_0, \dots, c_{n-1}) \in C\}$. Equivalently, $\underline{C}_I$ consists of all codewords of C with their i -th entries removed for each $i \in I$. The *shortening* of C at positions in I is the code $\overline{C}_I := \{(c_j)_{j \notin I} \mid (c_0, \dots, c_{n-1}) \in C, c_i = 0 \text{ for all } i \in I\}$. In other words, $\overline{C}_I$ consists of all codewords of C that vanish on the coordinates in I , with those coordinates removed.

2.2 Subfield subcodes

Given a code $C \subseteq \mathbb{F}^n$, the subfield subcode $C_{\mathbb{B}}$ is defined by $C_{\mathbb{B}} = C \cap \mathbb{B}^n$. If C has dimension k , then the dimension of $C_{\mathbb{B}}$ is lower-bounded by $n - (n-k)t$. Given a basis $\zeta = (\zeta_1, \dots, \zeta_t)$ of $\mathbb{F}$ over $\mathbb{B}$ and a matrix $M = (m_{ij}) \in \mathbb{F}^{u \times v}$, we write $M_{\mathbb{B}} \in \mathbb{B}^{ut \times v}$ to express M over the base field by expanding each entry column-wise. In other words, $M_{\mathbb{B}}$ denotes a matrix whose rows are

²We omit the minimum distance d , as it is not needed in the following.

indexed by (i, ℓ) and whose components are $m_{(i, \ell), j} = (m_{ij})_\ell$, where $(m_{ij})_\ell$ is the ℓ -th coordinate of m_{ij} in the basis ζ . If $H \in \mathbb{F}^{n \times (n-k)}$ is a parity-check matrix of C , then $H_{\mathbb{B}}$ is a parity-check matrix of $C_{\mathbb{B}}$, which is not necessarily of full rank.

2.3 Reed–Solomon and Generalized Reed–Solomon codes

Given $\alpha = (\alpha_0, \dots, \alpha_{n-1}) \in \mathbb{F}^n$, with $\alpha_i \neq \alpha_j$ for $i \neq j$, and $0 \leq k \leq n$, the Reed–Solomon code of length n associated with α , denoted by $\text{RS}[\alpha, k]$, is defined as $\text{RS}[\alpha, k] = \{(f(\alpha_0), \dots, f(\alpha_{n-1})) : f(X) \in \mathbb{F}[X], \deg f(X) < k\}$, which is an $[n, k]_{\mathbb{F}}$ -code. A *generalized Reed–Solomon code*, defined by α as above and by *column multipliers* $\mu = (\mu_0, \dots, \mu_{n-1}) \in (\mathbb{F} \setminus \{0\})^n$, denoted by $\text{GRS}[\alpha, \mu, k]$, is the code whose codewords are obtained from Reed–Solomon codewords by componentwise multiplication with the entries of μ :

$$\text{GRS}[\alpha, \mu, k] = \{(\mu_0 f(\alpha_0), \dots, \mu_{n-1} f(\alpha_{n-1})) : f(X) \in \mathbb{F}[X], \deg f(X) < k\}.$$

The dual code is $\text{GRS}[\alpha, \mu^\perp, n - k]$ for some $\mu^\perp = (\mu_0^\perp, \dots, \mu_{n-1}^\perp) \in (\mathbb{F} \setminus \{0\})^n$ [MS77, Thm 4, page 304].

2.4 LERS by Guruswami and Wootters

In the linear exact repair problem, the goal is to recover an erased code symbol using the minimal amount of information from the remaining code symbols. Without loss of generality, we assume that the first code symbol is erased. In [GW16], it is shown that, in certain settings, it suffices to download only partial information from each remaining code symbol in order to recover the erasure. More precisely, a symbol in $\mathbb{F}$ is decomposed into t sub-symbols over a subfield $\mathbb{B}$. Then, by performing local computation on these sub-symbols (linear in the case of [GW16]), one or more subfield elements can be used to recover the erased code symbol.

Definition 2. A equally distributed linear exact repair scheme (*LERS*) for an $[n, k]_{\mathbb{F}}$ -code C , with respect to a subfield $\mathbb{B} \subseteq \mathbb{F}$, and an integer $s \in \mathbb{N} \setminus \{0\}$ consists of

- $\mathbb{B}$ -linear maps $g_j: \mathbb{F} \rightarrow \mathbb{B}^s$ for $j \in \{1, \dots, n-1\}$,
- a $\mathbb{B}$ -linear reconstruction map $R: (\mathbb{B}^s)^{n-1} \rightarrow \mathbb{F}$,

such that for all $c = (c_0, \dots, c_{n-1}) \in C$ it holds $c_0 = R(g_1(c_1), \dots, g_{n-1}(c_{n-1}))$.

In the following, we exclusively focus on the case $s = 1$.

Definition 3 (Repair Bandwidth). The repair bandwidth b_0 is the total number of sub-symbols over $\mathbb{B}$ returned by the remaining code symbols at positions $1, \dots, n-1$, that is, $b_0 = \sum_{i=1}^{n-1} \dim_{\mathbb{B}}(\text{Im}(g_i))$.

Remark 4. The global repair bandwidth $b = \max_{i \in \{0, \dots, n-1\}} b_i$ is the worst-case number of sub-symbols required to repair an arbitrary erased code symbol, where the maximum is taken over all possible erasure positions.

The main result of Guruswami and Wootters establishes that, for full-length Reed–Solomon codes defined over a field $\mathbb{F}$, an erased codeword symbol can be recovered by downloading a single subfield element from each remaining share, resulting in a total download of $n - 1$ subfield elements.

Theorem 5 ([GW16]). Let $[\mathbb{F} : \mathbb{B}] = t$, $n = q^t$ and $k \leq (1 - \frac{1}{q})n$. Then the Reed–Solomon code $RS[\alpha, k]$ of length n , with support equal to the whole field $\mathbb{F}$, admits an equally distributed LERS over $\mathbb{B}$ with repair bandwidth $n - 1$.

2.5 Formulation with parity equations

The linear exact repair problem can be reformulated using parity equations. For this, a LERS can be constructed from a matrix H_t consisting of t codewords $h^{(1)}, \dots, h^{(t)} \in C^\perp$:

$$H_t = \begin{bmatrix} h^{(1)} \\ h^{(2)} \\ \vdots \\ h^{(t)} \end{bmatrix} = \begin{bmatrix} h_0^{(1)} & h_1^{(1)} & \dots & h_{n-1}^{(1)} \\ h_0^{(2)} & h_1^{(2)} & \dots & h_{n-1}^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ h_0^{(t)} & h_1^{(t)} & \dots & h_{n-1}^{(t)} \end{bmatrix}$$

satisfying

$$\dim_{\mathbb{B}}(\text{span}_{\mathbb{B}}(h_0^{(1)}, \dots, h_0^{(t)})) = t, \quad (1)$$

and

$$\dim_{\mathbb{B}}(\text{span}_{\mathbb{B}}(h_j^{(1)}, \dots, h_j^{(t)})) = 1 \quad \forall j \in \{1, \dots, n-1\}. \quad (2)$$

Theorem 6. Let C be a $[n, k]$ non-degenerated (see definition 7) linear code. Then the existence of a matrix H_t which satisfy (1) and (2) implies the existence of an equally distributed LERS for C with bandwidth $n - 1$.

Proof. We mimic the proof of [GW16]. By the definition of the dual code, for all $c \in C$ and $h \in C^\perp$, it holds $0 = \sum_{j \in \{0, \dots, n-1\}} c_j \cdot h_j$. Fix a basis $\zeta_1, \dots, \zeta_t$ of $\mathbb{F}$ over $\mathbb{B}$, and choose codewords $h^{(1)}, \dots, h^{(t)} \in C^\perp$ such that $h_0^{(i)} = \zeta_i$. Note that we can find such codewords since the dual code is $\mathbb{B}$ -linear and H_t satisfies condition (1). By condition (2), there exist elements $\beta_1, \dots, \beta_{n-1} \in \mathbb{F}$ such that $h_j^{(i)} \in \text{span}_{\mathbb{B}}(\beta_j)$ for all $j \in \{1, \dots, n-1\}$ and $i \in \{1, \dots, t\}$. Hence, for each $i \in \{1, \dots, t\}$, the codeword $h^{(i)} = (h_0^{(i)}, h_1^{(i)}, \dots, h_{n-1}^{(i)})$ can be written in the form $(\zeta_i, \lambda_1^{(i)} \cdot \beta_1, \dots, \lambda_{n-1}^{(i)} \cdot \beta_{n-1})$, where $\lambda_1^{(i)}, \dots, \lambda_{n-1}^{(i)} \in \mathbb{B}$.

For each $i \in \{1, \dots, t\}$, we have

$$\begin{aligned}
0 &= c_0 \cdot h_0^{(i)} + \sum_{j \in \{1, \dots, n-1\}} c_j \cdot h_j^{(i)} \\
0 &= c_0 \cdot \zeta_i + \sum_{j \in \{1, \dots, n-1\}} c_j \cdot \lambda_j^{(i)} \cdot \beta_j \\
-\zeta_i \cdot c_0 &= \sum_{j \in \{1, \dots, n-1\}} c_j \cdot \lambda_j^{(i)} \cdot \beta_j \\
-\text{Tr}(\zeta_i \cdot c_0) &= \sum_{j \in \{1, \dots, n-1\}} \text{Tr}(c_j \cdot \beta_j) \cdot \lambda_j^{(i)}.
\end{aligned}$$

The last equality follows from the $\mathbb{B}$ -linearity of the trace map (i.e., $\text{Tr}(a \cdot \lambda) = \lambda \cdot \text{Tr}(a)$ for $a \in \mathbb{F}$ and $\lambda \in \mathbb{B}$) and the fact that $\lambda_j^{(i)} \in \mathbb{B}$. Note that, after querying the $n - 1$ sub-symbols $\text{Tr}(c_j \cdot \beta_j)$, we can compute all t linear functions of c_0 of the form $\text{Tr}(\zeta_i \cdot c_0)$, since the coefficients $\lambda_j^{(i)}$ are known. Finally, let $\{\nu_1, \dots, \nu_t\} \subseteq \mathbb{F}$ be the dual basis of $\{\zeta_1, \dots, \zeta_t\}$, thus $\text{Tr}(\zeta_i \cdot \nu_\ell) = \delta_{i\ell}$ for all $i, \ell \in \{1, \dots, t\}$. Then the erased symbol c_0 can be obtained from the downloaded trace values using the identity

$$c_0 = \sum_{i=1}^t \text{Tr}(\zeta_i \cdot c_0) \nu_i.$$

Hence, exactly one $\mathbb{B}$ -sub-symbol is downloaded from each of the remaining $n - 1$ code symbols. Therefore, the repair bandwidth is $n - 1$, and the resulting equally distributed scheme is $\mathbb{B}$ -linear. Note that any choice of the basis of $\mathbb{F}$ over $\mathbb{B}$ yields a LERS. $\square$

3 Construction of LERS through subfield subcodes

We introduce the notation and definitions required in the sequel. Given an $[n, k]$ -code C , we write $\underline{C}$ for the *punctured code* at coordinate 0 (see Section 2.1 for the case $I = \{0\}$):

$$\underline{C} = \left\{ (c_1, \dots, c_{n-1}) \in \mathbb{F}^{n-1} \mid \text{there exists } c_0 \in \mathbb{F} \text{ such that } (c_0, \dots, c_{n-1}) \in C \right\}.$$

In other words, $\underline{C}$ is the projection of C onto the last $n - 1$ coordinates. Given a codeword $c \in C$, we write $c = (c_0, \underline{c})$, where $c_0 \in \mathbb{F}$ and $\underline{c} \in \underline{C}$.

Definition 7. A linear code C is said to be non-degenerate at position 0, if there exists a codeword in $C^\perp$ with a nonzero coordinate at position 0.

Definition 8 (C -extension). Given a code C that is non-degenerate at position 0, the C -extension map, denoted by $\phi_C: \underline{C} \rightarrow \mathbb{F}$, maps $\underline{c} \in \underline{C}$ to the unique element $c_0 \in \mathbb{F}$ such that $(c_0, \underline{c}) \in C$.

Note that a non-degenerate code is needed to ensure that the C -extension map is well defined. We write $\phi_C(c_1, \dots, c_{n-1}) = \sum_{j=1}^{n-1} \phi_j c_j$, and by a slight abuse of notation, we denote $\underline{\phi}_C = (\phi_1, \dots, \phi_{n-1})$. Observe that $(-1, \underline{\phi}_C) \in C^\perp$. To derive an explicit formula for $\phi_{C^\perp}$, note that if $C^\perp$ is non-degenerate, then we can write the generator matrix G of C , without loss of generality, such that its first row is of the form $(-1, g_1, \dots, g_{n-1})$, with $g_i \in \mathbb{F}$. It follows that the extension map $\phi_{C^\perp}$ is given, for $\underline{h} = (h_1, \dots, h_{n-1}) \in \underline{C}^\perp$, by

$$\phi_{C^\perp}(\underline{h}) = \sum_{i=1}^{n-1} g_i h_i. \quad (3)$$

Given a parity check matrix H of C , we can write H as $H = \begin{pmatrix} -1 & \underline{\phi}_C \\ 0 & \underline{H} \end{pmatrix}$, where $\underline{H}$ is a generating matrix of the shortening of $C^\perp$ at position 0.

3.1 Subfield subcode analysis for general linear codes

We consider a general construction where C is a non-degenerate (at position 0) $[n, k]_{\mathbb{F}}$ code. Our aim is to construct a matrix H_t from the generators of columns 1 to $n-1$. Suppose that there exists such a matrix H_t satisfying condition (1). Then consider $\beta = (\beta_1, \dots, \beta_{n-1}) \in \mathbb{F}^{n-1}$, where each β_j is a generator element of the j -th column of H_t . Our goal is to construct t distinct dual codewords $h \in C^\perp$ satisfying both (1) and (2). We first construct $\underline{h}$, the puncturing of h at position 0. We denote by $C_\beta^\perp$ the set of all vectors $\underline{h} \in \underline{C^\perp}$ that are $\mathbb{B}$ -collinear to β_j in position j . More precisely,

$$C_\beta^\perp := \left\{ \underline{h} = (h_1, \dots, h_{n-1}) \in \underline{C^\perp} : h_j \in \text{span}_{\mathbb{B}}(\beta_j) \ \forall j \in \{1, \dots, n-1\} \right\}.$$

Observe that $C_\beta^\perp$ is a $\mathbb{B}$ -linear code over the alphabet $\mathbb{F}$. Furthermore, for every $\underline{h} \in C_\beta^\perp$, there exist $\lambda_1, \dots, \lambda_{n-1} \in \mathbb{B}$ such that $h_j = \lambda_j \beta_j$ for $j = 1, \dots, n-1$. Define

$$\Lambda_\beta := \left\{ (\lambda_1, \dots, \lambda_{n-1}) \in \mathbb{B}^{n-1} \mid \exists \underline{h} \in C_\beta^\perp \text{ such that } h_j = \lambda_j \beta_j, \ \forall j \right\}.$$

Then $\Lambda_\beta \subseteq \mathbb{B}^{n-1}$ is a $\mathbb{B}$ -linear code over the alphabet $\mathbb{B}$. We consider that $\beta_j \neq 0$ for all $j \in \{1, \dots, n-1\}$, such that the inverse $\beta^{-1} = (\beta_1^{-1}, \dots, \beta_{n-1}^{-1})$ and the associated diagonal matrix $M_{\beta^{-1}} = \text{Diag}(\beta_1^{-1}, \dots, \beta_{n-1}^{-1})$ are well defined. Thus, Λ_β is the $\mathbb{B}$ -subfield subcode of $C_\beta^\perp M_{\beta^{-1}}$ i.e. $\Lambda_\beta = C_\beta^\perp M_{\beta^{-1}} \cap \mathbb{B}^{n-1}$.

Lemma 9. *The relationship $\Lambda_\beta = C_\beta^\perp M_{\beta^{-1}} \cap \mathbb{B}^{n-1}$ holds.*

Proof. Observe that $C_\beta^\perp \subseteq \mathbb{F}^{n-1}$ is $\mathbb{F}$ -linear, whereas $\Lambda_\beta \subseteq \mathbb{B}^{n-1}$ is $\mathbb{B}$ -linear. Let $\lambda \in \Lambda_\beta$, then there exists $\underline{h} \in C_\beta^\perp$ such that $h_j = \lambda_j \beta_j$

for all $j \in \{1, \dots, n-1\}$. Since $\beta_j \neq 0$ for all j , the diagonal matrix $M_{\beta^{-1}} = \text{Diag}(\beta_1^{-1}, \dots, \beta_{n-1}^{-1})$ is well defined and invertible. Hence,

$$\lambda = (\lambda_1, \dots, \lambda_{n-1}) = \left(\frac{h_1}{\beta_1}, \dots, \frac{h_{n-1}}{\beta_{n-1}} \right) = \underline{h} \cdot M_{\beta^{-1}} \in C_{\beta}^{\perp} M_{\beta^{-1}}.$$

where $\underline{h}$ is viewed as a row vector, and multiplication by $M_{\beta^{-1}}$ on the right corresponds to componentwise multiplication by the scalars β_j^{-1} . Therefore, $\lambda \in C_{\beta}^{\perp} M_{\beta^{-1}}$, which implies $\lambda \in C_{\beta}^{\perp} M_{\beta^{-1}} \cap \mathbb{B}^{n-1}$.

To prove the reverse inclusion, let $g \in C_{\beta}^{\perp} M_{\beta^{-1}} \cap \mathbb{B}^{n-1}$. Then there exists $\underline{h} \in C_{\beta}^{\perp}$ such that $g = (h_1, \dots, h_{n-1}) M_{\beta^{-1}}$. It follows,

$$g = (h_1, \dots, h_{n-1}) \text{Diag}((\beta_1^{-1}, \dots, \beta_{n-1}^{-1})) = (\lambda_1, \dots, \lambda_{n-1}) \in \Lambda_{\beta}.$$

We conclude, Λ_{β} is precisely the $\mathbb{B}$ -subfield subcode of the $\mathbb{F}$ -linear code $C_{\beta}^{\perp} M_{\beta^{-1}}$. $\square$

The lower bound on the dimension of the subfield subcode yields

$$\dim_{\mathbb{B}} \Lambda_{\beta} \geq (n-1) - (n-1 - (n-1-k))t, \quad (4)$$

since $\underline{(C^{\perp})}$ has $\mathbb{F}$ -dimension $n-1-k$ and length $n-1$. Simplifying the expression gives $\dim_{\mathbb{B}} \Lambda_{\beta} \geq (n-1) - kt$. In order for Λ_{β} to have positive dimension, it is therefore sufficient that $k < (n-1)/t$. We now turn to condition (1). Consider the $\mathbb{B}$ -linear $C^{\perp}$ -extension map:

$$\phi_{C^{\perp}}: \begin{cases} \underline{(C^{\perp})} \longrightarrow \mathbb{F} \\ \underline{h} \longmapsto \sum_{j=1}^{n-1} \phi_j h_j. \end{cases}$$

Let $\underline{h} \in C_{\beta}^{\perp}$, and write $h_j = \lambda_j \beta_j$, where $(\lambda_1, \dots, \lambda_{n-1}) \in \Lambda_{\beta}$. Then

$$\phi_{C^{\perp}}(\underline{h}) = \sum_{j=1}^{n-1} \phi_j h_j = \sum_{j=1}^{n-1} \phi_j \lambda_j \beta_j.$$

This induces a $\mathbb{B}$ -linear map

$$\psi: \begin{cases} \Lambda_{\beta} \longrightarrow \mathbb{F} \\ (\lambda_1, \dots, \lambda_{n-1}) \longmapsto \sum_{i=1}^{n-1} \phi_i \beta_i \lambda_i. \end{cases} \quad (5)$$

If the dimension of the image of ψ is t , then there exist $\underline{h}^{(1)}, \dots, \underline{h}^{(t)} \in C_{\beta}^{\perp}$, such that they extend to a basis $\zeta_1, \dots, \zeta_t$. Let $\ker(\psi)$ and $\text{Im}(\psi)$ denote the kernel and image of ψ , respectively. Then $\dim_{\mathbb{B}}(\ker(\psi)) + \dim_{\mathbb{B}}(\text{Im}(\psi)) =$

$\dim_{\mathbb{B}}(\Lambda_{\beta})$. In order to have $\dim_{\mathbb{B}}(\text{Im}(\psi)) = t$, it is necessary that $\dim_{\mathbb{B}}(\ker(\psi)) = \dim_{\mathbb{B}}(\Lambda_{\beta}) - t$. In particular, this implies $\dim_{\mathbb{B}}(\Lambda_{\beta}) \geq t$, which yields

$$k \leq \frac{n-1}{t} - 1. \quad (6)$$

In the following, the form of a parity-check matrix of Λ_{β} is discussed. Note that the parity-check matrix of $C^{\perp}$ is given by a generating matrix G of C . Suppose that G has the form $\begin{pmatrix} -1 & \underline{g} \\ 0 & \underline{G}' \end{pmatrix}$, where $\underline{g} \in \mathbb{F}^{1 \times (n-1)}$ and $\underline{G}' \in \mathbb{F}^{(k-1) \times (n-1)}$. Then $\underline{G}'$ is a parity-check matrix of $\underline{(C^{\perp})}$, and $\begin{pmatrix} \underline{g} \\ \underline{G}' \end{pmatrix}$ is the parity-check matrix of $\underline{(C^{\perp})} \cap \ker(\phi_{C^{\perp}})$. Hence, for $\underline{h} \in \underline{(C^{\perp})}$, with $h_j = \lambda_j \beta_j$, and $M_{\beta} = \text{Diag}(\beta_1, \dots, \beta_{n-1})$ it holds

$$G' \underline{h}^T = G' (\underline{\lambda} M_{\beta})^T = (G' M_{\beta}) \underline{\lambda}^T.$$

Consequently, a (not necessarily full-rank) parity-check matrix of Λ_{β} is given by $(G' M_{\beta})_{\mathbb{B}} \in \mathbb{B}^{t(k-1) \times (n-1)}$. Similarly, $\ker(\psi)$ is the right kernel of the matrix

$$\begin{pmatrix} (\underline{g} M_{\beta})_{\mathbb{B}} \\ (G' M_{\beta})_{\mathbb{B}} \end{pmatrix}. \quad (7)$$

It remains to determine β such that the rank of matrix in (7) is exactly t larger than the rank of $(G' M_{\beta})_{\mathbb{B}}$. To summarize, the subfield subcode construction succeeds if condition (6) is fulfilled and a suitable β is determined.

3.2 Simulations

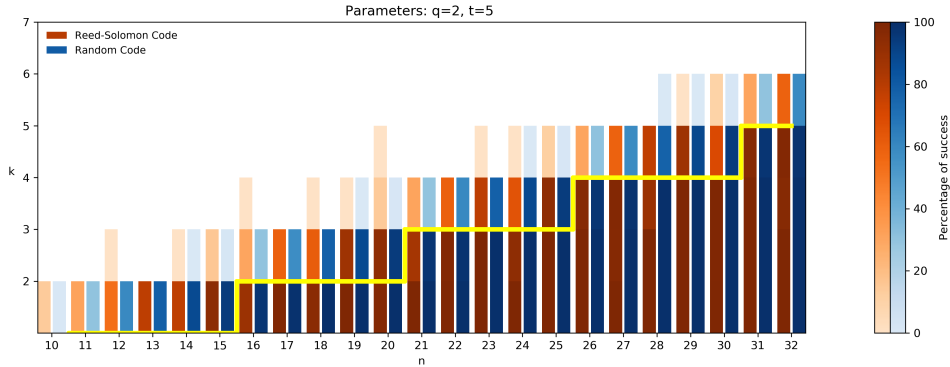


Figure 1: Percentage of success (for 10000 trials for each pair (n, k)) for $q = 2$ and $t = 5$. Here success means that the rank of the matrix in (7) is exactly t larger than the rank of $(G' M_{\beta})_{\mathbb{B}}$. The threshold for k is indicated in yellow.

In general, it is very hard to mathematically determine the rank of a matrix M_β given M . Consequently, we rely on numerical simulations. The empirical results of these simulations for fixed $q = 2$ and $t = 5$ are shown in Figure 1 for both Reed–Solomon codes and random codes. In each of the 10000 trials for each pair (n, k) , the elements $\beta_1, \dots, \beta_{n-1} \in \mathbb{F} \setminus \{0\}$ were chosen independently and uniformly at random for each pair (n, k) . For each such choice of $\beta_1, \dots, \beta_{n-1}$, the matrix in (7) is constructed, and we verify whether its $\mathbb{B}$ -rank is exactly t greater than the rank of $(G'M_\beta)_\mathbb{B}$. However, this equality does not hold in general. The results indicate that the rank of the matrix in (7) often exceeds the rank of $(G'M_\beta)_\mathbb{B}$ by exactly t , and this behavior becomes more pronounced as k decreases. In particular, for sufficiently small k , the success rate reaches 100%. The threshold for k , given by the upper bound $k \leq (n - 1)/t - 1$, is indicated in the figure (highlighted in yellow). We observe that the success rate often attains 100% whenever k satisfies this bound, providing empirical support for our subfield subcode theory. Moreover, the observed behavior for Reed–Solomon codes closely matches that for randomly generated codes, with only a marginally higher success rate for Reed–Solomon codes. This difference may be due to sampling variability, while the asymptotic behavior could be identical. Additional simulations covering further parameter choices exhibit similar behavior.

3.3 Comparison to existing explicit LERS

In this subsection, we compare the advantages and limitations of our explicit subfield subcode construction of a LERS with the Reed–Solomon LERS proposed by Guruswami and Wootters. The scheme in [GW16] applies only to full-length Reed–Solomon codes and requires the dimension to satisfy $k \leq q^t - q^{t-1}$. In contrast, our construction applies to any non-degenerate $[n, k]$ code over $\mathbb{F}$ and guarantees an LERS whenever $k \leq (n - 1)/t - 1$.

4 Conclusion

Inspired by the LERS of [GW16], the main contribution of this work is the development of a new subfield subcode-based construction of a LERS that applies to any non-degenerate code over an extension field of length n and dimension $k \leq (n - 1)/t - 1$. Our LERS covers a new range of parameters beyond previously known schemes by allowing $n < q^t$. Moreover, our construction imposes no structural restrictions on n or the code while, by design, achieving low communication costs—specifically, a repair bandwidth which does not exceed $n - 1$. As an open research question, it remains whether a theoretical probability corresponding to the statistical experiment presented in Section 3.2 can be derived.

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